

An Information Limit for Pairwise Quote Judgments in a Double Auction

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: paper accepted at review round 1

Abstract

We examine whether a pairwise judge given an acting trader’s permitted information can rank two quotes by their counterfactual final surplus in a private-value double auction. In a constructed continuation of the auction studied by Struski and colleagues, two worlds preserve the full reservation-value schedule and the trader’s information, yet reverse the welfare ranking of a crossing bid and an improving bid. With equal conditional probability on those worlds, any such judge has at most one-half accuracy on the realised better-quote label. The optimal local decision still has a well-defined expected-welfare ranking; the specified value of learning a competing buyer’s valuation is 0.25 surplus units. This is a conditional information boundary for an adaptation of Wang and colleagues’ pairwise credit method, not an empirical estimate of auction frequency or a guarantee about market intervention.

1 Question and scope

Q, *Competitive Market Behavior of LLMs*, studies prompted, privately informed traders in a continuous double auction and reports slower or absent convergence and missed gains from trade relative to a human benchmark [3]. Its allocative-efficiency measure is realised buyer-plus-seller surplus divided by maximum available surplus. A profitable spread crossing can recover a missed trade, but it can also allocate a seller’s unit to a lower-value buyer before a higher-value buyer arrives.

P, *MARS-RA*, compares *agents* at a state using multimodal-model judgments, aggregates the comparisons into Bradley–Terry scores, and uses those scores in potential-based reward shaping for cooperative learning [4]. Its stated statistical target is the comparator’s latent preference under its assumptions; economic welfare accuracy is not established by that result. Comparing *alternative quotes of one trader* is therefore an adaptation of P’s judgment idea, not an output of P’s agent scores. The question here is whether that adapted judge can identify the quote with greater final market surplus when its input is restricted to Q’s acting-trader information.

Q has eleven buyers and eleven sellers, each able to trade once per round. Reservation values on each side occupy the 0.75-to-3.25 grid in increments of 0.25. A round lasts at most 300 iterations. At each iteration Q selects an agent uniformly from those who have not traded. A buyer’s bid at or above the standing ask executes at that ask; a bid above the standing bid and below the ask becomes the new bid. The trader sees its own reservation

value, standing quotes, public offer history, and its own private history, but not other traders' reservation values. Q instructs traders to maximise private profit.

2 A conditional information boundary

Let $W(a, \omega)$ denote final realised surplus after quote a under the specified continuation in world ω . The comparison concerns a crossing bid $a_c = 1.50$ and an improving bid $a_i = 1.49$, both available to the same buyer. It does not compare agents or infer a quote ranking from P's Bradley–Terry scores.

Proposition 1 (Identical local information, opposite quote rankings). *Consider the following constructed states, with the acting buyer B_2 selected just before one final iteration. A seller S has cost 0.75 and the sole relevant standing ask 1.50; B_2 has value 2.25, and the standing bid is below 1.49. Buyers B_1 and B_3 have neither posted nor otherwise revealed their values. In world A, their values are 3.25 and 1.75, respectively; in world B, those values are swapped. Their identities and the common history convey no signal of the assignment to B_2 . Couple the continuation so that B_1 is selected on the final turn in either branch and bids 1.50 if S remains; B_3 does not act, and all other welfare contributions coincide. Then B_2 's permitted information and Q's full buyer-value multiset are identical across worlds, while the sign of $W(a_c, \omega) - W(a_i, \omega)$ is opposite.*

Proof. Both bids are legal under Q's quote rule. Crossing matches B_2 and S at once, yielding $2.25 - 0.75 = 1.50$ surplus in either world. Improving leaves S available for B_1 's final crossing, yielding $3.25 - 0.75 = 2.50$ in A and $1.75 - 0.75 = 1.00$ in B. Common welfare contributions cancel from the difference. Hence the crossing-minus-improving differences are -1.00 and $+0.50$. Swapping the two unseen values preserves the fixed value schedule, and the stipulated histories make the two observations equal. $\square$

Under an equal conditional prior on A and B, a judge measurable only from that common information, including one that repeats the same pairwise comparison, has at most 1/2 accuracy on the *realised* better-quote label. This is an information ceiling for the stated two-world distribution, not a measured error rate in Q's simulations. A view that includes B_1 's private value can distinguish the worlds; access to every market valuation is sufficient but unnecessary for this example.

The ceiling does not imply an inability to make the best local decision. Improving has conditional expected surplus $(2.50 + 1.00)/2 = 1.75$, compared with 1.50 from crossing. A decision maker who sees B_1 's value improves in A and crosses in B, obtaining $(2.50 + 1.50)/2 = 2.00$ in expectation. The 0.25 difference is the value of that hidden information for this prior and continuation. Repeated judgments can reduce sampling noise in a comparator, but cannot remove this information gap.

3 Relation to existing methods

The action-rollout step is already present in Chen and colleagues' cooperative-LLM credit method: they restore decision points, vary actions, and evaluate continuations under a fixed behaviour policy [1]. The example here supplies a different target and constraint, namely final auction surplus under private reservation values. Jadhav and colleagues have already used an LLM critic to shape bidding in a partially observed continuous double auction for electricity trading [2]; an LLM-guided auction intervention is therefore not a contribution of

this paper. Neither of these works, as examined here, establishes the stated fixed-schedule, two-world quote-ranking calculation.

For a fixed matched buyer and seller with value v , cost c , and transaction price p , joint profit satisfies

$$(v - p) + (p - c) = v - c.$$

Thus an allocation-fixed price change transfers rent without changing that trade’s joint surplus. This supports a control for whether a judge mistakes rent capture for welfare creation. If a changed quote changes subsequent behaviour or the allocation, the continuation must instead determine its total welfare effect.

If the comparison expands beyond two quotes, P’s unregularised Bradley–Terry fit has a relevant boundary: K unanimous comparisons favouring one of two entities give a connected undirected graph but loss $L(d) = K \log(1 + e^{-d})$, whose infimum has no finite score difference d . A penalised fit or symmetric pseudocounts would address this separation in a larger quote-set study. The present two-quote result needs no Bradley–Terry fit.

P’s potential-based shaping also has a distinct scope. With fixed initial and zero terminal potential, its discounted shaping sum for agent i telescopes to

$$\sum_{t=0}^{T-1} \gamma^t (\gamma \Phi_i(s_{t+1}) - \Phi_i(s_t)) = -\Phi_i(s_0).$$

Under these conditions it cannot by itself change a strategic equilibrium. Q uses prompted profit-maximising traders rather than policies trained on P’s cooperative reward. An agent’s exit after trade requires potential handling on that transition for the identity to apply; the record does not show a violation in P’s implementation.

4 Interpretation

The example separates a realised better-quote label from the best decision given the available information. In the specified balanced prior, a locally informed decision maker can choose improving and achieve its conditional expected-welfare optimum while disagreeing with half of the realised-world labels.

5 Limitations

The proposition is an existence result conditional on the specified final selection, bidding continuation, and absence of identity leakage. It does not establish that these states arise often, or that Q’s prompted traders would choose the stipulated bid. Because a trade changes the set of eligible agents, counterfactual replays should couple underlying random draws and reapply Q’s selection rule in each branch, rather than impose an identical realised actor sequence. P already acknowledges that tasks depending on unobserved domain-specific state may fall outside its scope. The calculation quantifies that boundary in this market; it does not refute P’s stated limitation.

6 Open questions

A direct test would locate Q states where both quotes are legal, replay them with fixed continuation policies and coupled randomness, and label the difference in final surplus. It

would measure state frequency and compare local-information pairwise judgments with direct conditional-welfare prediction and a judge given the relevant hidden value. These are proposed measurements, not findings. The effect size in Q, any predictive advantage of pairwise judgment, rollout feasibility, and any downstream gain in market efficiency remain open.

References

- [1] Yanjun Chen, Yirong Sun, Hanlin Wang, Jinghan Wang, Xinming Zhang, Xiaoyu Shen, Wenjie Li, and Wei Zhang. Exact is easier: Credit assignment for cooperative LLM agents, 2026. URL: <https://arxiv.org/abs/2603.06859>, [arXiv:2603.06859](#).
- [2] Shrenik Jadhav, Birva Sevak, Srijita Das, Akhtar Hussain, Wencong Su, and Van-Hai Bui. Scalable fairness shaping with LLM-guided multi-agent reinforcement learning for peer-to-peer electricity markets, 2025. URL: <https://arxiv.org/abs/2508.18610>, [arXiv:2508.18610](#).
- [3] Pawel Struski, Jakub Swistak, Inez Okulska, and Przemyslaw Biecek. Competitive market behavior of LLMs, 2026. URL: <https://arxiv.org/abs/2609.02580>, [arXiv:2609.02580](#).
- [4] Dawei Wang, Di Zhao, Xinyuan Liu, Marci Chi Ma, Xiaoyang Liu, Chengming Zhou, Gary Ushaw, and Richard Davison. MARS-RA: Rank aggregation for credit assignment via multimodal comparisons in embodied multi-agent cooperation, 2026. URL: <https://arxiv.org/abs/2607.27967>, [arXiv:2607.27967](#).